Digital Twin-Driven Deep RL for Energy-Aware CNC Milling

Dynamic Cutting Parameter Optimization

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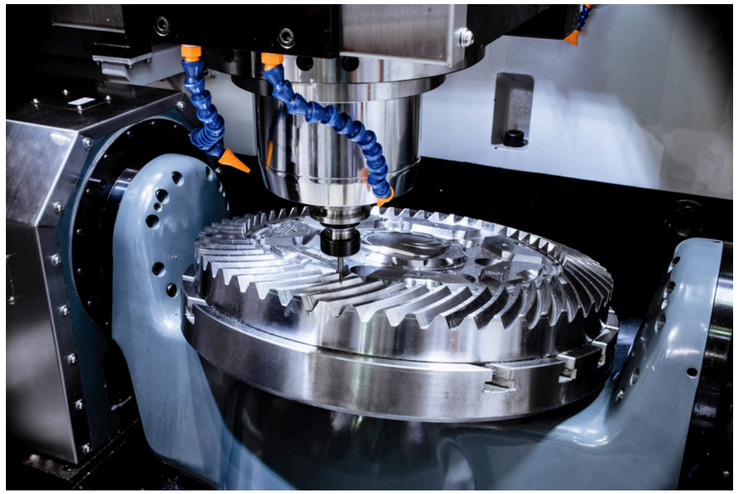
Problem Statement

Challenges in CNC Machining:

- Optimal cutting parameters (speed, feed, depth) are critical to balance throughput and quality.
- Traditional meta-heuristic algorithms optimize these statically.
- Real-world multi-pass operations experience dynamic tool degradation (wear and stochastic breakage).

Modern Manufacturing Needs:

- Requires energy-efficient operations.
- Static approaches cannot adapt to rapidly changing tool conditions on the fly.



Source : https://www.fastems.com/cnc-machining/

Proposed Methodology : Deep RL

MDP Formulation

$$S_t = (VB_i, \kappa, n_{i-1}, f_{i-1}, ap_{i-1}, MRR_{i-1}, Ra_{i-1}, B_i) \in [0,1]^8$$

- VB_i (Tool Wear): The current physical flank wear of the cutting tool (measured in mm).
- κ (Remaining Allowance): The total depth of material that still needs to be removed to finish the workpiece (mm)
- n_{i-1} (Previous Spindle Speed): The rotational speed of the spindle used during the last cutting pass (RPM).
- f_{i-1} (Previous Feed Rate): The speed at which the tool advanced against the workpiece during the last pass (mm/rev)
- ap_{i-1} (Previous Depth of Cut): The depth of the cut made during the previous pass (measured in mm).
- MRR_{i-1} (Previous Material Removal Rate): The volume of material removed per minute during the last pass, indicating cutting throughput (measured in mm^3/min).
- Ra_{i-1} (Previous Surface Roughness): The calculated surface finish quality achieved on the workpiece after the last pass (measured in µm).
- B_i (Tool Breakage State): A boolean safety flag (1 if the tool has catastrophically broken, 0 if it is still intact).

Action Space

The agent outputs a continuous 3-dimensional vector defining the cutting parameters for the next pass.

$$A_t = (n, f_r, a_p)$$

- n (Spindle Speed): The rotational speed of the cutting tool (Bounded: 2000 to 8000 RPM).
- f_r (Feed Rate): The distance the tool advances per revolution (Bounded: 0.05 to 0.35 mm/r).
- a_p (Depth of Cut): The depth the tool plunges into the material (Bounded: 0.1 to 2.5 mm).

Reward Function

$$\mathbf{r_t = r_{VB} + r_B + r_{Ra} + r_{MRR} + r_{Energy} + r_{HV} + r_{step}}$$

$$r_{VB} = \begin{cases} 0, & VB \leq VB_{max} \\ -50, & \text{otherwise} \end{cases}$$

$$r_B = \begin{cases} 0, & \text{Tool intact} \\ -100, & \text{Tool broken} \end{cases}$$

$$r_{Ra} = \begin{cases} 0, & Ra \leq Ra_{max} \\ -50, & \text{otherwise} \end{cases}$$

$$r_{MRR} = 0.5 \left(\frac{MRR}{MRR_{max}} \right)$$

$$r_{Energy} = -w_2 \left(\frac{E}{E_{max}} \right)$$

$$r_{HV} = \left(\frac{\max(0, T_{ref} - T_{total})}{T_{ref}} \right) \times \left(\frac{\max(0, Ra_{ref} - Ra)}{Ra_{ref}} \right) \times 100$$

$$r_{step} = -0.1$$

Reinforcement Learning Algorithm

The chosen algorithm is **Proximal Policy Optimization (PPO)**.

- PPO is an on-policy algorithm that strikes an excellent balance between sample efficiency and ease of tuning.
- Given the continuous nature of the action space and the complexity of the multi-objective delayed Hypervolume rewards.
- PPO ensures monotonic policy improvements and stable gradient updates, which are essential for converging toward a robust, constraint-compliant machining policy while mitigating the risk of Knowledge degradation and Unbounded policy divergence

Digital Twin Physics (Cutting & Throughput)

- Cutting Speed (m/min): $v_c = \frac{\pi \cdot D \cdot n}{1000}$
- Feed per Minute (mm/min): $f = n \cdot f_r$
- Cutting Time (min): $t_{cut} = \frac{L_{cut}}{f}$
- Material Removal Rate (mm³/min): $MRR = v_c \cdot a_p \cdot f_r$

D = diameter of the tool
n = spindle speed
fr = feed rate
L_cut = cutting length of the workpiece
a_p = axial depth of the cut

Digital Twin Physics (Degradation & Energy)

Tool Wear Degradation:

Modeled via the Taylor-extended equation to capture accelerating wear as cutting speed and feed rate increase.

$$\Delta VB = C_{wear} \cdot v_c^{\alpha} \cdot f_r^{\beta} \cdot a_p^{\gamma} \cdot t_{cut}$$

- ΔVB = incremental tool flank wear (wear increase)
- Cwear = tool-wear coefficient
- vc = cutting speed
- fr = feed rate
- ap = axial depth of cut
- tcut = cutting time
- α,β,γ = empirical exponents representing the sensitivity of tool wear to cutting speed, feed rate, and depth of cut

DRL-based optimization for machining process: Cutting parameters dynamic optimization considering tool condition degradation, J. Manuf. Processes, 2025

Digital Twin Physics (Degradation & Energy)

Surface Roughness Surrogate Model:

An empirical regression formula linking surface finish directly to feed rate, cutting speed, and current tool wear.

$$Ra = k_1 f_r^2 + k_2 VB + \frac{k_3}{v_c}$$

Machining parameter optimization for a batch milling system using multi-task deep reinforcement learning, J. Manuf. Systems, 2024

- Ra = surface roughness
- fr = feed rate
- VB = current tool flank wear
- vc = cutting speed
- k1,k2,k3 = empirically fitted regression coefficients

Energy Consumption (Novel addition)

$$E = (P_{idle} + k_c \cdot MRR) \cdot t_{cut}$$

- E = total machining energy consumption
- Pidle = machine power consumed when idle
- kc = specific cutting energy coefficient
- MRR = material removal rate
- tcut = cutting time